

DSS Application Design Document

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1 Overview

This is a design document for a distributed storage system (DSS). It includes the message formats for the implemented commands, the time-space diagrams, the data structures and design decisions, the port numbers, the screenshots, the video demo link and timestamps.

2 Message Formats for Implemented Commands

This milestone uses a common message envelope for all requests and responses. Individual commands define their own bodies and semantics.

2.1 Envelope Format (Common to All Messages)

Envelope

```
{
  "version": 1,
  "message_id": "<role>-<pid>-<timestamp_ms>-<counter>",
  "message_type": "<command_or_response>",
  "status_code": "SUCCESS | FAILURE (optional on requests)",
  "reason": "Optional explanation on FAILURE",
  "body": { /* command-specific fields */ }
}
```

Message ID Format

Pattern: <role>-<pid>-<timestamp_ms>-<counter> (e.g., user1-4123-1732665001123-00042).

Role: user name or disk name (e.g., manager, user1, diskA)

PID: process ID of the sender

Timestamp: current time in milliseconds

Counter: incremented each time this process sends a message

Purpose: match requests and responses, enable retries on UDP, deduplicate requests, and simplify logging.

2.2 Command: register_user

Purpose

Register a user process with the manager.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user",
  "body": {
    "user_name": "User1",
    "ipv4_address": "192.0.2.10",
    "management_port": 2101,
    "command_port": 2102
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665001123-00001",
  "message_type": "register_user_response",
  "status_code": "SUCCESS",
  "reason": null
}
```

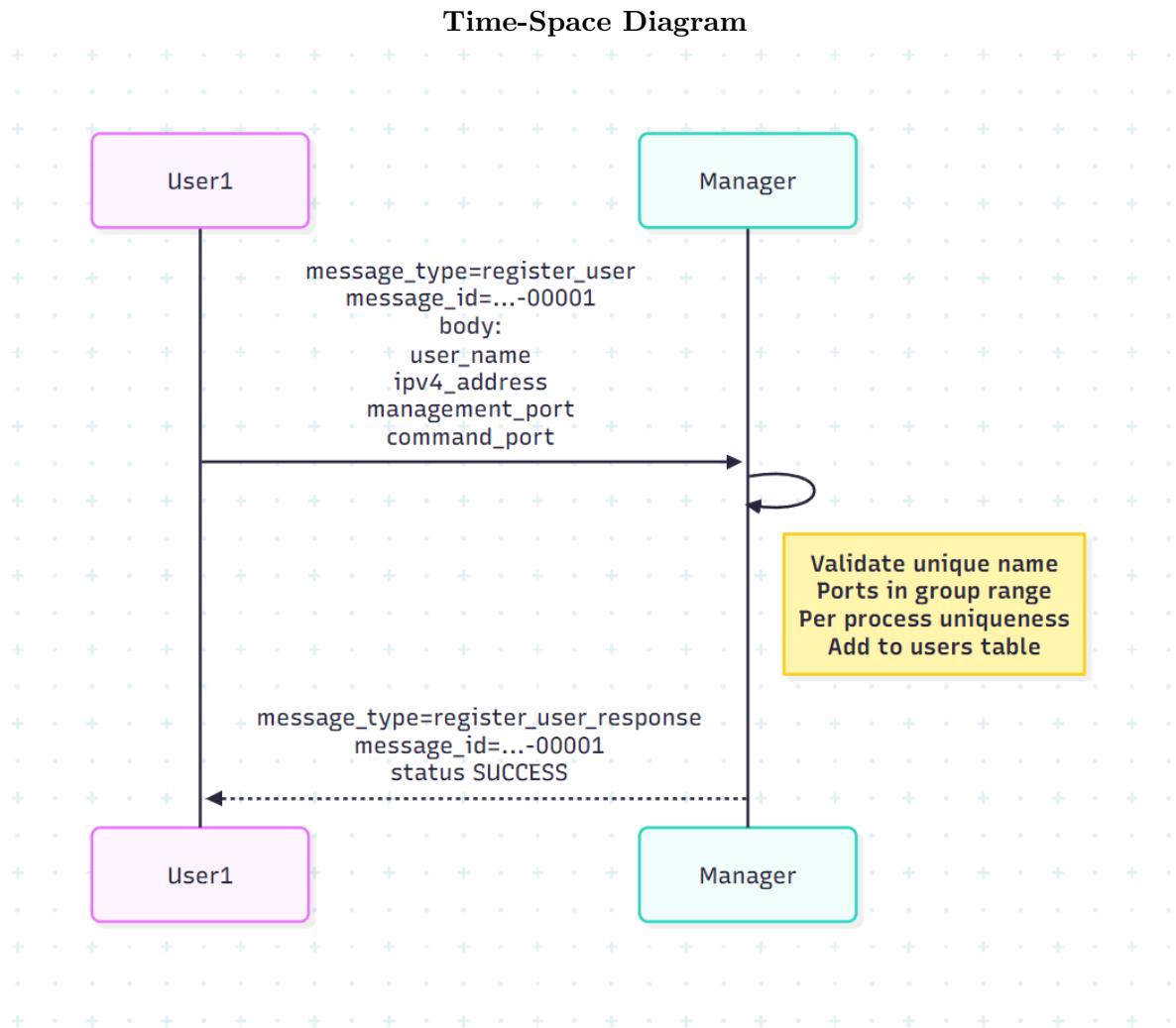


Figure 1: Time-Space diagram for `register_user`

2.3 Command: register_disk

Purpose

Register a disk process with the manager.

Request

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk",
  "body": {
    "disk_name": "DiskA",
    "ipv4_address": "192.0.2.11",
    "management_port": 2201,
    "command_port": 2202
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "diskA-6123-1732665001567-00001",
  "message_type": "register_disk_response",
  "status_code": "SUCCESS"
}
```

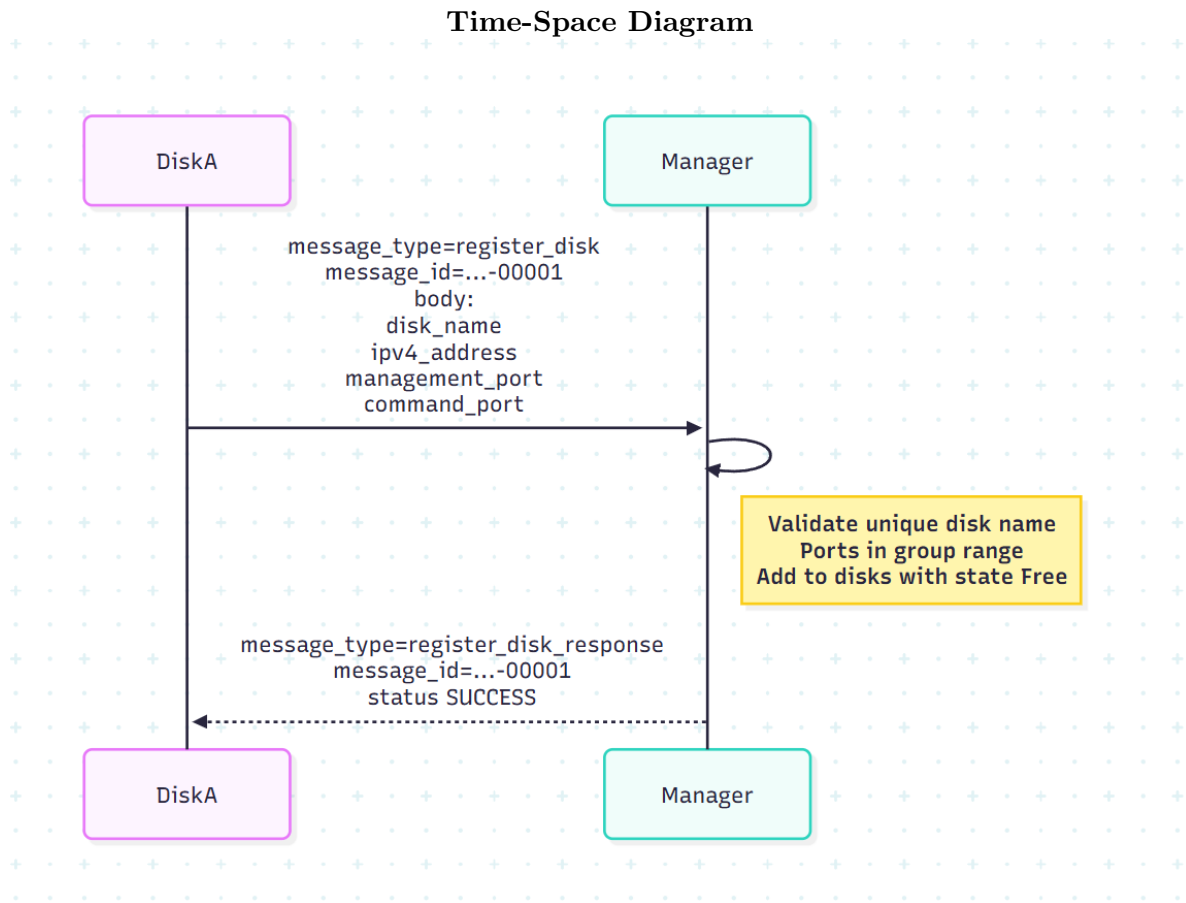


Figure 2: Time-Space diagram for `register_disk`

2.4 Command: configure_dss

Purpose

Configure a DSS from available disks.

Request

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss",
  "body": {
    "dss_name": "DSS1",
    "number_of_disks": 3,
    "striping_unit_bytes": 1024
  }
}
```

Response

```
{
  "version": 1,
  "message_id": "user1-4123-1732665002001-00002",
  "message_type": "configure_dss_response",
  "status_code": "SUCCESS",
  "body": {
    "dss_name": "DSS1",
    "disk_order": ["DiskA", "DiskB", "DiskC"],
    "striping_unit_bytes": 1024
  }
}
```

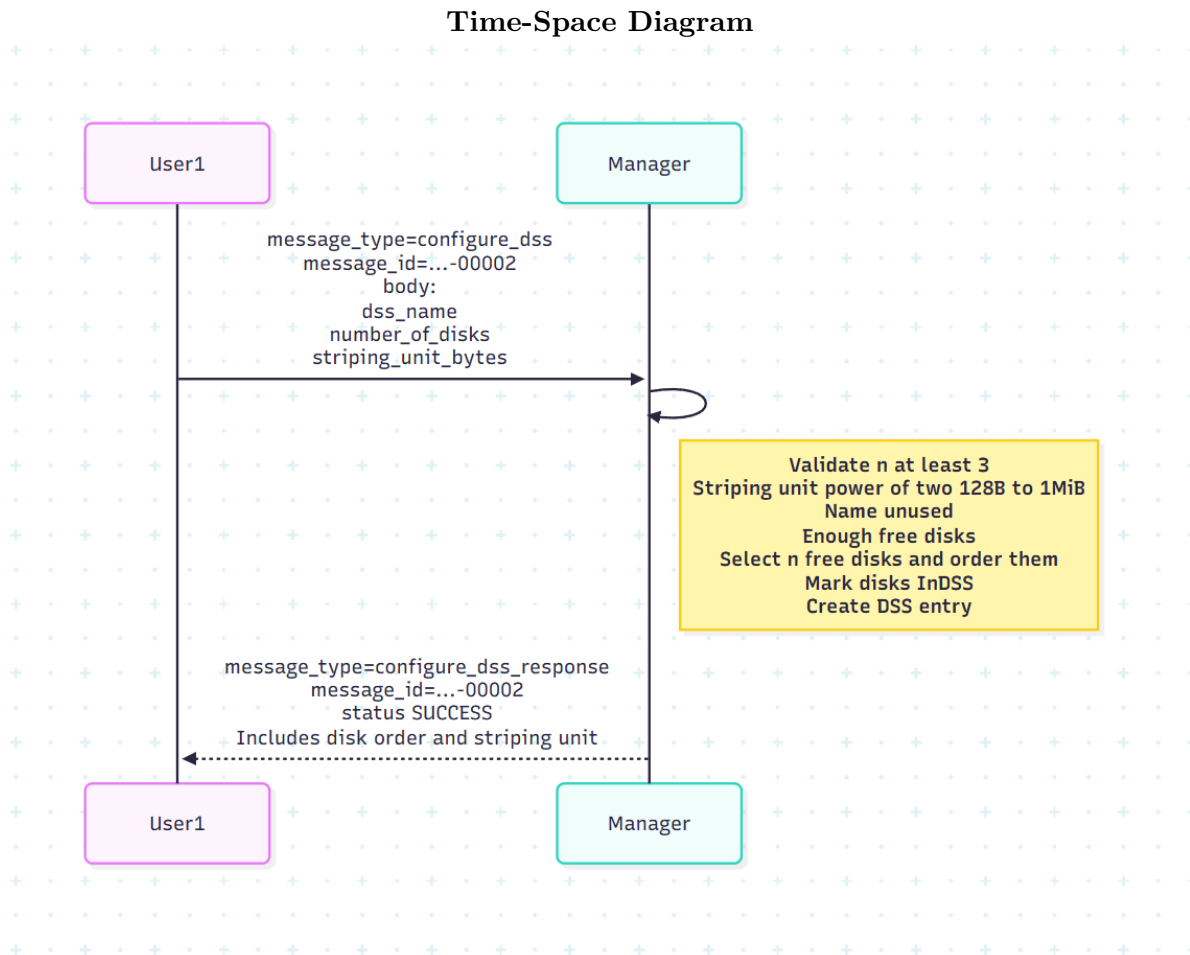


Figure 3: Time-Space diagram for `configure_dss`

2.5 Command: deregister_user

Purpose

Deregister a user process from the manager.

Request

```
{  
  "version": 1,  
  "message_id": "user1-4123-1732665002500-00003",  
  "message_type": "deregister_user",  
  "body": {  
    "user_name": "User1"  
  }  
}
```

Response

```
{  
  "version": 1,  
  "message_id": "user1-4123-1732665002500-00003",  
  "message_type": "deregister_user_response",  
  "status_code": "SUCCESS"  
}
```

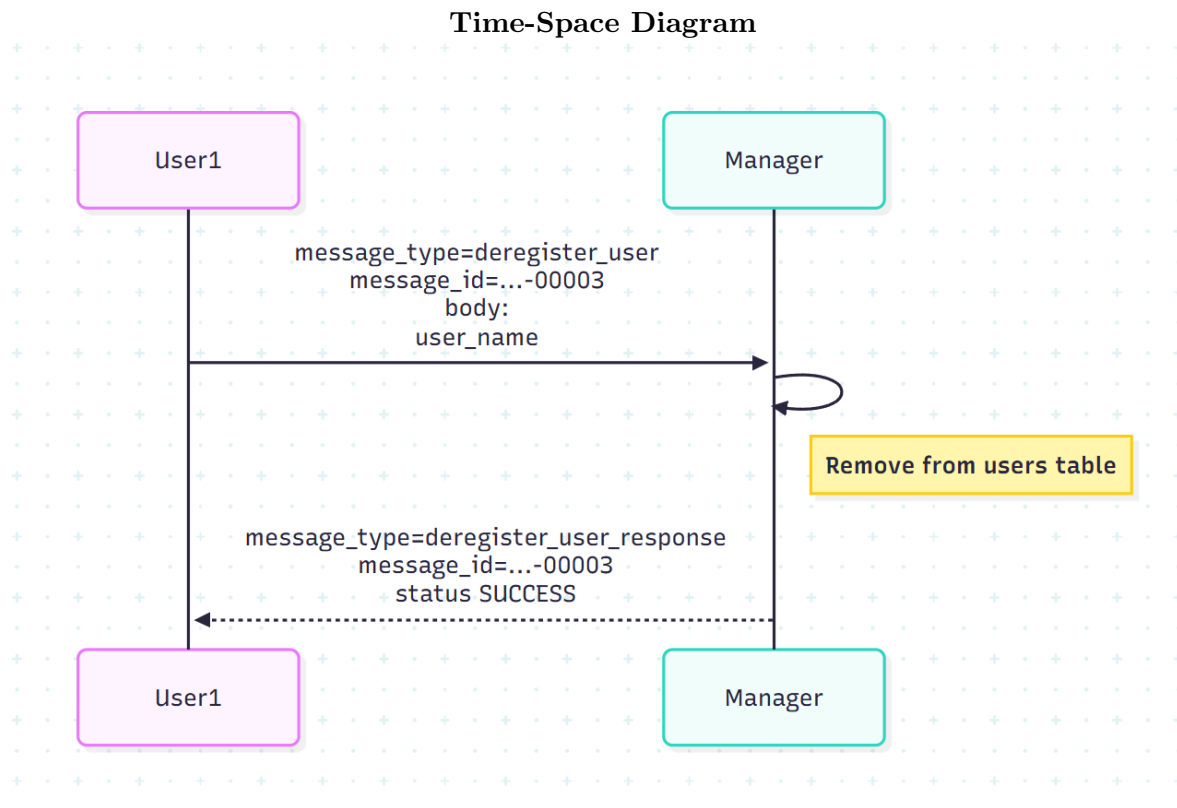


Figure 4: Time-Space diagram for `deregister_user`

2.6 Command: deregister_disk

Purpose

Deregister a disk process from the manager.

Request

```
{  
  "version": 1,  
  "message_id": "diskA-6123-1732665002700-00004",  
  "message_type": "deregister_disk",  
  "body": {  
    "disk_name": "DiskA"  
  }  
}
```

Response

```
{  
  "version": 1,  
  "message_id": "diskA-6123-1732665002700-00004",  
  "message_type": "deregister_disk_response",  
  "status_code": "SUCCESS"  
}
```

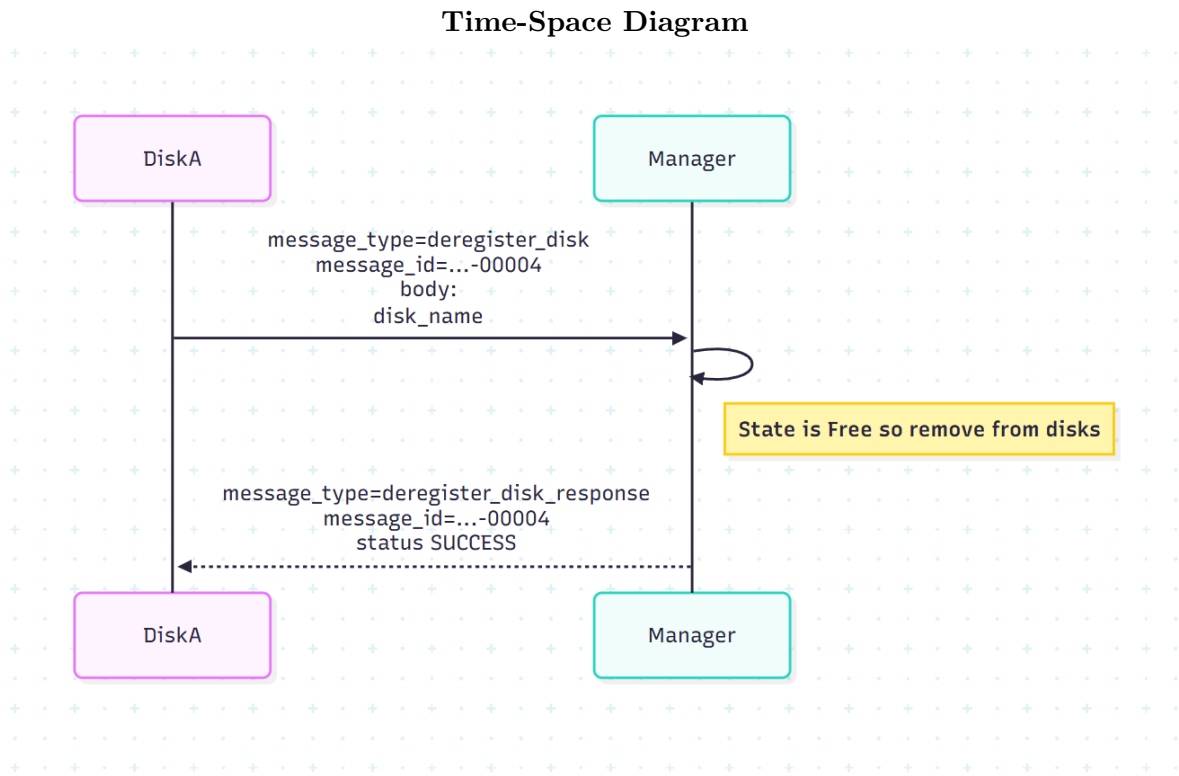


Figure 5: Time-Space diagram for `deregister_disk`

3 Design Decisions

Key design decisions include: - Using a common message envelope for all messages. - Using a shared message ID pattern for all messages. - Using standardized response/request message formats for all commands.

4 Port Numbers

Because I am Group 198, the port numbers are 21200–21299 (inclusive).

5 Screenshots

Provide the following screenshots (place files under `screenshots/`):

- i. Output of: `git log --pretty=format:"%h - %an, %ad (Commit) - %cd (Author)"` (TBD)
- ii. Entire commit history of the GitHub repository (TBD)
- iii. Output of: `git reflog` (TBD)
- iv. Output of: `git fsck` (TBD)

Note I didn't include the entire git history due to it being paywalled by GitHub Plus.

TBD: Insert screenshot of git log command output

Figure 6: `git log` output

TBD: Insert screenshot of full commit history

Figure 7: Repository commit history

TBD: Insert screenshot of `git reflog` output

Figure 8: `git reflog` output

TBD: Insert screenshot of `git fsck` output

Figure 9: `git fsck` output

6 Video Demo Link and Timestamps (3(a)–3(f))

Accessible Link TBD

Timestamps TBD.

1. 3(a): TBD
2. 3(b): TBD
3. 3(c): TBD
4. 3(d): TBD

5. 3(e): TBD
6. 3(f): TBD

7 Extended Design for Full Project

For the next version of this project, I'll need to add commands, add threading, and record another video demo.